

Performance characteristics, continued

Expected values
Eleven serum samples from a variety of different species were tested in the assay. Values ranged from 0.78 mg/dL to 1.45 mg/dL with an average of 1.00 mg/dL.

Sensitivity
The detection limit was determined at two standard deviations from the zero along the standard curve. **Sensitivity was determined as 0.081 mg/dL.**

Cross reactivity and interferences
A serum sample was spiked with varying concentrations of bilirubin and tested in the assay. Bilirubin level in normal serum is between 0.2 mg/dL and 1.0 mg/dL. The unspiked sample read 0.86 mg/dL. No significant change to the measured creatinine level was seen up to an additional 1.0 mg/dL of bilirubin.

Limited product warranty

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REF

Catalog Number

LOT

Batch code

Temperature limitation

Use by

Manufacturer

Consult instructions for use

Caution, consult accompanying documents

Manufacturer: Life Technologies Corporation | 7335 Executive Way | Frederick, MD 21704 | USA

The information in this guide is subject to change without notice.

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Serum Creatinine Colorimetric Detection Kit

Catalog Number EIASCR (192 tests), EIASCRX10 (10 x 192 tests)

Rev B.0

Note: For safety and biohazard guidelines, see the “Safety” appendix in the *ELISA Technical Guide* (Pub. no. MAN0006706). Read the Safety Data Sheets (SDSs) and follow the handling instructions. Wear appropriate protective eyewear, clothing, and gloves.

Product description

The Serum Creatinine Colorimetric Detection Kit is designed to detect and quantify the level of creatinine in mammalian serum and plasma samples. The assay was validated for human, mouse, rabbit, rat, and sheep serum and EDTA and heparin plasma samples. Creatinine (2-amino-1-methyl-5H-imidazol-4-one) is a metabolite of phosphocreatine (p-creatine), and molecule used as a store for high-energy phosphate and can be utilized by tissues to produce ATP. Creatinine comes from either the diet or synthesis from the amino acids arginine, glycine, and methionine. This occurs in the kidneys and liver, although other organ systems may be involved and species-specific differences may exist.

Serum creatinine is another metabolic waste product removed from the blood by the kidneys, but it exhibits a steady rate of elimination and can be used to normalize urea nitrogen values.

Contents and storage

Kit and components are shipped at –20°C. Upon receipt, store the kit at –20°C. Once open, store the kit at 4°C and use within 2 weeks.

Components	Quantity (192 tests)	Quantity (10 x 192 tests)
Creatinine Standard: 100 mg/dL	100 µL	10 x 100 µL
Assay Diluent: special diluent for use in the serum kit. Allow to warm completely to room temperature prior to use.	6 mL	10 x 6 mL
Clear Half-Area 96-well plate	2 plates	20 plates
Creatinine Reagent	20 mL	10 x 20 mL

Materials required but not supplied

- Distilled or deionized water
- Microtiter plate reader with software capable of measurement at or near 490 nm
- Calibrated adjustable precision pipettes and glass tubes for diluting solution

Procedural guidelines

IMPORTANT! Reagents are lot-specific. Do not mix or interchange different reagent lots from various kit lots.

Sample preparation guidelines

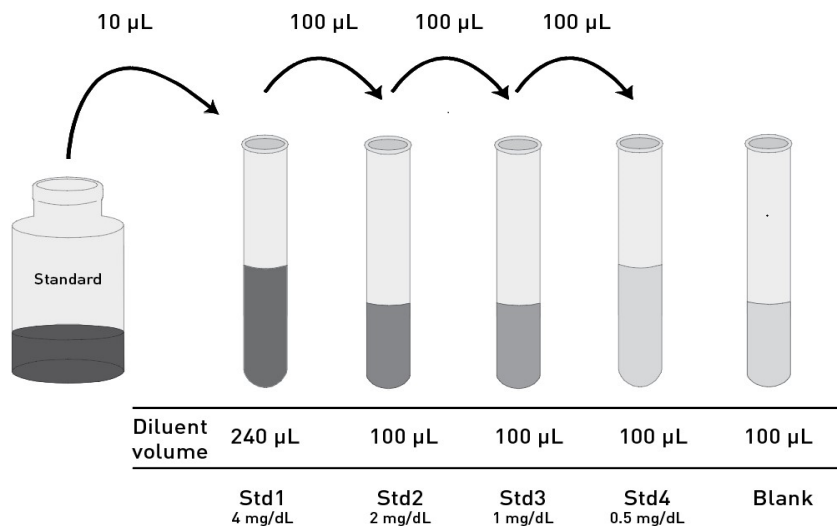
- Collect samples in pyrogen/endotoxin-free tubes.
- Freeze samples after collection if samples will not be tested immediately. Avoid multiple freeze-thaw cycles of frozen samples. Thaw completely and mix well (do not vortex) prior to analysis.
- Avoid the use of hemolyzed or lipemic sera.
- If large amounts of particulate matter are present in the sample, centrifuge or filter sample prior to analysis.
- All samples should be centrifuged for 15 minutes at 14,000 rpm in an Eppendorf-type centrifuge prior to running in the assay.

Dilute standards

Note: Use glass tubes for diluting standards.

Allow the kit reagents to come to room temperature for 30 minutes.

1. Add 10 μL Creatinine Standard to one tube containing 240 μL water and label as 4 mg/dL.
2. Add 100 μL water to each of 3 tubes labeled as follows: 2, 1, and 0.5 mg/dL.
3. Make serial dilutions of the standard as described below in the dilution diagram. Mix thoroughly between steps. Water is used as a sample blank of 0 mg/dL.
4. Use the standards within 2 hours of preparation.



Perform Assay

IMPORTANT! Perform a standard curve with each assay.

Allow all components to reach room temperature before use. Mix all liquid reagents prior to use.

1. Prior to starting the assay, ensure the plate reader is set to read absorbance at 490 nm, all reagents are at room temperature, and have a timer set to 30 minutes. It is recommended to pipette each sample, blank, and standard in duplicate for improved accuracy and precision of measured concentrations.
2. Add 25 μL of samples, blank (water), or standards into duplicate wells.
3. Add 25 μL of Assay Diluent to all wells. If bubbles are present, gently tap the plate to remove them.
4. Add 100 μL of the Creatinine Reagent to all wells. Start the 30-minute timer immediately after addition as this step initiates the reaction.
5. When the timer reaches 1 minute, read the plate at 490 nm (T-1min).
6. When the timer reaches 30 minutes, read the plate a second time at 490 nm (T-30min).
7. Calculate the average of the Optical Density (AvgOD) of each duplicate sample, blank, and standard for both plate readings.
8. Subtract the AvgOD of T-1min from the AvgOD of T-30min.
9. Plot the difference in these ODs against the concentration of each standard, then generate a linear regression for the standard curve. Use the equation $y=mx+b$ (where y = average difference in ODs, x = creatinine concentration, m = slope, b = intercept) to calculate the concentration of unknown samples.

Performance characteristics

Standard curve (example)

The following data were obtained for the various standards over the range of 0.5-4 mg/dL creatinine.

Standard Creatinine (mg/dL)	Optical Density (490 nm)
4	0.384
2	0.187
1	0.090
0.5	0.047

Intra-assay precision

Samples were assayed in replicates of 20 to determine precision within an assay.

Parameters	Sample 1	Sample 2	Sample 3
Mean [mg/dL]	0.99	1.50	3.82
CV (%)	7	6.3	4.5

CV = Coefficient of Variation

Inter-assay precision

Samples were assayed in duplicates in 19 assay runs by four operators to determine precision between assays.

Parameters	Sample 1	Sample 2	Sample 3
Mean [mg/dL]	0.91	1.26	3.51
CV (%)	9.6	7.3	8.0

CV = Coefficient of Variation